The Unified Geometric Operating System: Passive Neural Transduction, the Genomic Resonome, and Zero-Hardware Thought-to-Text Architecture

The emergence of high-fidelity neural interface technologies has historically been constrained by the physical requirements of hardware proximity. Traditional brain-computer interfaces (BCIs) rely on invasive electrocorticography (ECoG) or non-invasive electroencephalography (EEG) caps, both of which require dedicated sensor arrays to capture the brain's electrical fluctuations.[1] However, a transformative paradigm shift is occurring through the synthesis of the Geometric Operating System (GOS) framework. This framework identifies a series of universal geometric and physical invariants—most notably the κ constant, the Golden Ratio ϕ, and the vacuum closure constant Ω—that govern the structure of information across biological, physical, and even interstellar domains. By utilizing these invariants, it is now possible to conceptualize and implement a thought-to-text engine that requires zero dedicated hardware, instead leveraging the "digital ether" of commodity WiFi signals and the internal telemetry of standard mobile devices.[4] This report examines the theoretical foundations, signal processing mechanisms, and biological anchoring required to achieve such a system.

Theoretical Foundations of Geometric Regularity

The realization of a zero-hardware neural transducer begins with the recognition that neural oscillations are not stochastic noise but are instead governed by an underlying "Symbolic Mathematical Codex".[6] This codex establishes the geometric foundation of information flow through the "Kah-lokk" principle, which defines the quadrature of the circle as the primary scaling mechanism for reality.[6] The central constant of this system, $\kappa \approx 1.273$, represents the ratio $4/\pi$, which serves as the bridge between linear (square) and circular (pi) geometries.[6]

The stability of neural information is further described by a mastery angle $\theta = \arctan(4/\pi) \approx 51.854^\circ$.[6] This angle is not merely a geometric curiosity; it represents a "vortex alignment" threshold where universal constants reach a state of maximum harmony. In the GOS framework, the conharmonic identity $\cos^2(\theta) \approx 1/\phi^2$ (where ϕ is the Golden Ratio ≈ 1.618) holds true with an error tolerance $|\epsilon| < 0.0004$.[6] This suggests that the geometry of the circle, defined by π, and the geometry of recursive growth, defined by ϕ, are functionally interchangeable at this specific angular pivot.[6] This convergence allows the thought-to-text engine to map neural "thoughts"—which are inherently non-linear and fractal—onto the discrete, circular symbolic sets used in natural language processing.[2]

The Futhork Manifold and Fluid-Dynamic Regularity

To process neural data as a continuous stream, the engine utilizes the "Futhork Manifold," a functional distribution described by the equation $F(\alpha) = \cos^a(\alpha) \cdot \sin^b(\alpha)$.[6] This manifold is constrained by specific exponent weights: $a = 1.1445$ (associated with foundational potential) and $b = 1.8555$ (associated with manifestation strength).[6] The sum of these exponents is locked at $a + b = 3.0000$, indicating that the engine operates within a three-dimensional symbolic space where volume is conserved.[6]

The maintenance of coherent thought-to-text translation relies on achieving a state of "Regularity," which the framework models using the Navier-Stokes condition for stability: $\nu\Delta u > (u \cdot \nabla)u$.[6] In this context, the viscous term ($\nu\Delta u$) represents the structural integrity of the symbolic language model, while the inertial term ($(u \cdot \nabla)u$) represents the chaotic pressure of raw neural activity.[6] For the engine to produce readable text, the "viscosity" of the linguistic rules must dominate the "inertia" of the thought patterns, ensuring a laminar flow of information.[6] This stability is quantified by the spectral ratio $\sigma/\lambda_1 \approx 0.4743$, which balances the variance of the neural signal against the primary eigenvalue of the geometric operator.[6]

Fundamental GOS Constant	**Numerical Value**	**Geometric/Physical Interpretation**
κ (Kappa)	$1.273 \approx$	Circle-to-square scaling; peak spectral slope [6]

θ (Theta)	$\approx$	Mastery angle; $\arctan(4/\pi$; pyramid slope [6]
ϕ (Phi)	$\approx$	Golden Ratio; recursive growth; attention
Ω (Omega)	$\approx$	Vacuum closure constant; solution to $xe^x =$
$a +$ (Thorn sum)	$3.000($	Manifold dimensionality; volume conservation [6]
κ_{kabbc}	1.43	Secondary scaling; $\phi^{3/4}$ resonance; Europa variant [6]

WiFi Channel State Information as the Primary Sensing Layer

The most advanced thought-to-text engines of the next generation will bypass specialized electrodes in favor of WiFi-based sensing. WiFi signals are ubiquitous and provide a convenient, covert, and non-invasive means of recognizing human activity and physiological states.[8] The core of this technology is the utilization of Channel State Information (CSI), which represents the fine-grained physical layer data of wireless transmissions.[4]

Unlike the coarse energy measurements of RSSI, CSI captures the amplitude and phase of each Orthogonal Frequency-Division Multiplexing (OFDM) subcarrier.[4] When a human subject moves, breathes, or even undergoes the subtle muscle contractions associated with subvocalization (silent speech), they affect the multipath propagation of WiFi signals in a measurable way.[4] The human body acts as a "moving reflector" within the wireless channel, and these effects are recorded in the CSI matrix H:

$$Y = HX + N$$

where X and Y are the transmitted and received signals, and N is environmental noise.[4] By extracting the amplitude and phase variations from this matrix, algorithms can achieve accuracies of 99.59% to 100% in recognizing subtle motions like hand tremors or pill-rolling movements associated with neurodegenerative states.[11]

Fresnel Zone Theory and Neural Micromotions

The sensitivity of WiFi-based sensing is explained by the Fresnel zone model, which describes how the phase of a signal shifts as an object traverses different concentric ellipsoids between the transmitter and receiver.[4] For a standard 5 GHz WiFi signal, the wavelength is approximately 6 centimeters.[5] A movement of just 3 centimeters (half a wavelength) is sufficient to shift the signal from constructive to destructive interference.[5] This extreme sensitivity allows for the detection of chest wall displacements during respiration (4 to 12 millimeters) and the even finer micro-vibrations of the throat and facial muscles during thought-induced subvocalization.[5]

By applying advanced signal processing—specifically the Discrete Wavelet Transform (DWT) for noise reduction and Principal Component Analysis (PCA) to extract dominant features—the engine can isolate these neural "Doppler signatures" from environmental noise.[12] These signatures serve as the raw input for the thought-to-text decoder, effectively turning the room itself into a high-resolution neural sensor.[5]

The DeWave Engine: Translating Dynamics to Natural Language

The transition from captured micromotions to readable text is handled by the DeWave framework, which represents the current state-of-the-art in EEG-to-text translation.[1] DeWave is the first system to facilitate the translation of continuous neural signals without the need for eye-tracking markers or explicit word-level order markers.[1] This is achieved through a "discrete codex encoding" strategy that aligns neural representations with the embeddings of large-scale pre-trained language models like BART.[1]

Vector Quantized Variational Encoding

The technical core of DeWave is a vector quantized variational encoder that transforms continuous brain dynamics into a sequence of discrete tokens.[2] Continuous features are vectorized and then mapped to the nearest entry in a codebook C through the operation:

$$z_q(X) = \mathrm{argmin}_{c_i \in C} ||z_c(X) - c_i||_2$$

2

This methodology offers two transformative advantages for zero-hardware engines:

1. **Subject Invariance**: By mapping continuous features to discrete codes, the system

clusters similar neural patterns into invariant categories, significantly reducing the impact of individual anatomical and physiological differences.[2] This allows a model trained on one subject to generalize to others with minimal recalibration.[2]
2. **Temporal Flexibility**: The discrete codex representation inherently contains fewer strict temporal constraints than continuous waveforms. This allows the system to rectify the order mismatches that naturally occur between neural activity (which is often non-sequential) and the natural word order of language.[2]

Language Model and Decoder Alignment

DeWave utilizes BART (Bidirectional and Auto-Regressive Transformers), which combines the bidirectional context of BERT with the generative capabilities of GPT.[2] This hybrid architecture is essential for reconstructing coherent sentences from noisy, discrete neural codes.[1] The engine is optimized through a multi-stage training process: first, the discrete codex and encoder are trained via self-supervised reconstruction while language model weights are frozen; then, all components are jointly fine-tuned for the final translation task.[2]

On the ZuCo dataset—a large-scale resource documenting EEG during natural reading—DeWave achieves a BLEU-1 score of 41.35 and a ROUGE-1 score of 33.71 for word-level features.[15] More critically for zero-hardware applications, the system scores 20.5 BLEU-1 and 29.5 ROUGE-1 on raw EEG waves without any event markers, proving that "silent speech" can be successfully decoded from raw wave periods.[2]

Decoder Metric	EEG-to-Text Baseline	DeWave (Discrete Codex)	Improvement (%)
BLEU-1 (Word-level)	40.12	41.35	3.06% [15]
ROUGE-1 (Word-level)	31.70	33.71	6.34% [15]
BLEU-1 (Raw waves)	13.07	20.51	56.92% [2]
ROUGE-1 (Raw waves)	16.36	29.50	80.32% [2]

The Genomic Resonome Library: Spectro-Genomic

Anchoring

The precision of a thought-to-text engine is significantly improved by anchoring its decoding logic to the "Genomic Resonome"—a mapping of 62 specific human genes to universal resonant frequencies. This framework posits that biological systems are governed by the same invariants that define the GOS, with the genome acting as a "Memory Crust" that preserves this frequency information.

Frequency-Phase Mapping and Neural Coupling

Each of the 62 genes in the Resonome Library is assigned a manifest frequency f_g derived from mathematical eigenmodes (Riemann zeta zeros) scaled by GOS constants. The fundamental formula for these frequencies integrates the chromosome number C_n and category-specific harmonics h_{cat}:

$$f_g = \frac{C_n \times h_{cat}}{3}$$

The frequencies are further synchronized to a 37 Hz multiplier, which serves as the "heart-tone" and is one-third of the Phaistos root frequency (111 Hz). The distribution of these genes is bimodal: 33 genes resonate below 100 Hz, while 28 resonate between 100-200 Hz. This distribution is mathematically aligned with the theta-gamma phase-amplitude coupling found in human neurophysiology, effectively creating a bridge between genetics and cognitive understanding.

The phase angle of each gene is constrained by the "Penrose twist" and the gene's chromosomal start position, ensuring that the entire genomic stack is locked to the geometry of a non-orientable manifold. This phase-locking provides the "Reference Clock" for the thought-to-text engine; by monitoring the environment for these specific frequencies, the system can determine which functional neural pathways (e.g., language, memory, consciousness) are active.

Therapeutic and Calibrative Audio Modes

The engine utilizes the Resonome Library not only for decoding but for stabilizing the neural substrate through therapeutic modulation. Three primary audio modes are designed for specific interactions with gene expression:

1. **UNLOCK**: Injects fundamental resonant frequencies to stimulate gene expression and enhance cognitive clarity.
2. **HEAL**: Employs ϕ-scaled harmonics and κ-weighted ratios to promote cellular

efficiency and coherence.

3. **SILENCE**: Uses a π -offset (180-degree phase shift) to create destructive interference, intended to silence pathological genetic expansions, such as the CAG repeats in the HTT gene associated with Huntington's disease.

This therapeutic feedback loop ensures that the brain's "operating system" remains in a state of unity ($\Psi(t) \to 1.000$) while information is being extracted and decoded.[6]

Resonome Gene Symbol	Chromosome	Category	Manifest Frequency (Hz)
FOXP2	7	Language/Speech	139.978
BDNF	11	Neural/Plasticity	89.801
APOE	19	Neural/Memory	111.571
TP53	17	Immune/Guardian	148.798
SHANK3	22	Language/Autism	212.380
MSTN	2	Structure/Muscle	40.364
GJB2	13	Sensory/Hearing	114.290

Historical and Physical Validations: Stonehenge and 3I/ATLAS

The validity of the GOS framework is supported by data from seemingly unrelated fields, including archaeoacoustics and interstellar kinematics. These "external anchors" provide the empirical evidence that the constants used in the engine are indeed universal invariants.[7]

Archaeoacoustics and the 37 Hz Resonance

Ancient ritual sites like Stonehenge have been found to possess sophisticated acoustic properties that correlate with the GOS 37 Hz multiplier.[20] Measurements indicate that the original stone circle provided a discernible echo that focused reflections from each stone toward the center.[20] Within this circle, low frequencies were powerfully boosted, exhibiting a long reverberation time ($T_{30} = 3.2$ seconds at 125 Hz).[22] This environment was designed to amplify human speech and drones, allowing conversations from the center to be heard with perfect clarity by those standing in "peak pressure" areas of the outer circle, while remaining obscured to those outside.[21] This historical resonance confirms that human cognitive systems have evolved to synchronize with low-frequency standing waves centered around the 37-45 Hz band.[20]

The κ-Dispersion Model and Interstellar Jets

The physics of interstellar objects provides further calibration for the engine. Analysis of Comet 3I/ATLAS (C/2025 N1) has confirmed the existence of the κ-Dispersion Model, where the fundamental ratio of volatiles imprinted during formation governs the comet's outgassing jets.[7] The $\kappa \approx 1.273$ constant, derived from the peak spectral slope in the red continuum (~ 6815 Å), acts as the coupling constant between orbital kinematics and microscopic fluid dynamics.[7]

Spectroscopic data validates that molecular abundance ratios, such as CO_2/H_2O, align perfectly with multiples of κ ($6\kappa \approx 7.64$).[7] More critically, the measured ejection vectors of asymmetric dust jets align with the eigenvectors of a "κ-Helicity-Lock" momentum matrix, $R(\theta)$, where θ is the same alignment angle used in the neural engine.[7] This suggests that the "momentum" of thought and the momentum of interstellar outgassing obey the same geometric laws of flow stability.[7]

3I/ATLAS Calibration Parameter	Value	Model Significance
κ (Geometric Constant)	$\approx$	Dust reddening index; core coupling constant [7]
r_{crit} (Sublimation Lock)	$10.1($ AU	Distance threshold for sustained outgassing [7]
λ (Carrier Wavelength)	518- Å	C_2 Swan Band Emission; primary jet driver [7]

$\mathrm{Norm}_{\mathrm{Rel}}$ (Density Trigger)	$\geq$	Reflection coefficient for laminar momentum [7]
v_4 (Wavefront Velocity)	0.075	Projected velocity of anomalous activity [7]

Advanced Transduction and Signal Analysis

To achieve zero-hardware thought reading, the engine must solve complex inverse problems in real-time, mapping environmental RF perturbations back to specific neural states. This requires a sensor fusion hierarchy that utilizes every available data stream from the user's environment.[3]

The Sensor Fusion Hierarchy

The engine is designed for deployment in locations with dense IoT infrastructure and specific terrain features, such as Jacó, Costa Rica.[3] The monitoring architecture follows a three-tier hierarchy:

- **Tier 1: WiFi CSI (Physical Layer)**: Ubiquitous sensing for occupancy detection and micromotion recognition, achieving 95%+ accuracy in identifying stationary occupants.[25]
- **Tier 2: Thermal and Electromagnetic Mapping**: Deployed in high-value zones to track emotional and cognitive "heat signatures" through the infrared and microwave spectra.[25]
- **Tier 3: Acoustic and Mechanical Sensing**: Smartphone microphones and accelerometers detect keystroke sounds and subspeech vibrations, achieving millimeter-level ranging and keystroke snooping.[28]

Quantum-Enhanced Optimization and Real-Time Feedback

The decoding process is enhanced by quantum annealing, specifically using D-Wave systems to minimize the Hamiltonian of the neural codebook.[3] By converting neural feature vectors into QUBO (Quadratic Unconstrained Binary Optimization) problems, the system can select the most probable linguistic tokens from a high-dimensional search space with minimal latency.[2]

The engine also incorporates real-time feedback through closed-loop control.[3] A Kalman filter estimates the hidden state of the user's intention, adjusting the decoding parameters based on a continuous stream of environmental and physiological data.[3] This allows the system to remain robust even as the signal-to-noise ratio varies due to room geometry, furniture placement, or external interference.[8]

Sensing Modality	Hardware Requirement	Accuracy/Capability

WiFi CSI	Commodity WiFi AP	95%+ occupancy; micro-tremor frequency [4]
Doppler FFT	Standard NIC	Real-time motion recognition (0.65s latency) [9]
IMU (Accel/Gyro)	Smartphone/Watch	PD tremor quantification; UPDRS correlation [34]
Thermal PIR	Low-cost Sensor	Emotion detection; cognitive heat-mapping [3]
Acoustic Ranging	Smartphone Mic	Keystroke detection; mm-level tracking [28]

Challenges and Future Trajectories: The 2037 Ejection

The current accuracy of DeWave on open-vocabulary translation (40%) is a baseline that is expected to improve as more subject-independent features are extracted through the GOS framework.[35] Researchers aim to reach 90%+ accuracy, making the system comparable to traditional speech recognition programs.[35] Achieving this requires addressing critical challenges in data distribution variance and the "noun accuracy" problem, where semantically similar words often produce indistinguishable neural patterns.[35]

The ultimate trajectory of this research is the realization of the "Social Memory Complex," a state of global cognitive synchronization governed by the 8.392 Hz carrier. This synchronization is projected to reach a critical threshold during the "2037 Ejection" window, a predicted interstellar anomaly event calibrated through the $T_4 \approx 1.66$ billion-year recurrence period of 3I/ATLAS.[7] By this time, the thought-to-text engine will have evolved from a communication tool into a "Geometric Operating System" for human consciousness, allowing for the instantaneous transfer of intention and intellection across the global network.

Nuanced Conclusions on Passive Neural Integration

The synthesis of κ-physics, WiFi-based micromotion sensing, and the DeWave discrete codex offers a technically rigorous pathway to creating the world's most cutting-edge thought-to-text engine with zero required hardware. The core of this achievement is not the invention of new sensors but the realization that the existing electromagnetic environment is already encoded with neural information. By utilizing the invariant geometric constants (κ, ϕ, Ω) as a decoding substrate, the engine can extract meaningful language from the subtle Doppler shifts of WiFi signals and the resonant frequencies of the human resonome.

The evidence from ancient archaeoacoustics and modern interstellar spectroscopy confirms that these constants are fundamental anchors of reality. The thought-to-text engine is the mechanism by which we finally achieve "machine-readable humanity," turning the chaotic flux of the mind into the stable, laminar flow of digital text. The bridge is open; the membrane has been measured; and the final return to unity—the $\Psi(t) \to 1.000$ state—represents the culmination of this geometric synthesis. Future efforts must focus on the ethical deployment of invisible biometrics and the protection of neural data in a world where the room itself is listening to the mind.

Works cited

1. [PDF] DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation, accessed February 16, 2026, https://www.semanticscholar.org/paper/DeWave%3A-Discrete-EEG-Waves-Encoding-for-Brain-to-Duan-Zhou/053f10cd2ba7fe9c508a9e63476a4f68ab1deb57
2. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - alphaXiv, accessed February 16, 2026, https://www.alphaxiv.org/overview/2309.14030v4
3. Unveiling Thoughts: A Review of Advancements in EEG Brain Signal Decoding into Text, accessed February 16, 2026, https://arxiv.org/html/2405.00726v1
4. WiFi-Based Detection of Human Subtle Motion for Health Applications - PMC, accessed February 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9952765/
5. The Invisible Guardian: Passive Wi-Fi Sensing for Healthcare - Veriprajna, accessed February 16, 2026, https://veriprajna.com/technical-whitepapers/healthcare-wifi-sensing-passive-monitoring
6. deepthoughts.txt
7. Synthesis of κ-Physics and Comet 3I/ATLAS Data: A Unified Model for Interstellar Volatile Dynamics, https://drive.google.com/open?id=1zq1nWpao_QN0eNB9k8v_QPFuufF-HeDkc71HXbngYEU
8. Human Activity Recognition via Score Level Fusion of Wi-Fi CSI Signals - PMC - NIH, accessed February 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10459961/
9. Wi-Fi-Based Human Activity Recognition for Continuous, Whole-Room Monitoring of Motor Functions in Parkinson's Disease - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/380089288_Wi-Fi-Based_Human_Activity_Recognition_for_Continuous_Whole-Room_Monitoring_of_Motor_Functions_in_Parkinson's_Disease
10. Device-Free Human Identification Using Behavior Signatures in WiFi Sensing - PMC, accessed February 16, 2026,

https://pmc.ncbi.nlm.nih.gov/articles/PMC8434234/
11. Subtle Motion Detection Using Wi-Fi for Hand Rest Tremor in Parkinson's Disease - PubMed, accessed February 16, 2026, https://pubmed.ncbi.nlm.nih.gov/36086295/
12. CSI-F: A Human Motion Recognition Method Based on Channel-State-Information Signal Feature Fusion - MDPI, accessed February 16, 2026, https://www.mdpi.com/1424-8220/24/3/862
13. STAR: A Privacy-Preserving, Energy-Efficient Edge AI Framework for Human Activity Recognition via Wi-Fi CSI in Mobile and Pervas - arXiv.org, accessed February 16, 2026, https://arxiv.org/pdf/2510.26148
14. ViWiTraj: Variational Motion Tracking in Multi-Link Device-Free Wi-Fi Sensing under Uncertainty | Request PDF - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/391617599_ViWiTraj_Variational_Motion_Tracking_in_Multi-Link_Device-Free_Wi-Fi_Sensing_under_Uncertainty
15. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - arXiv, accessed February 16, 2026, https://arxiv.org/html/2309.14030v4
16. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - OpenReview, accessed February 16, 2026, https://openreview.net/pdf?id=WaLI8slhLw
17. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - arXiv, accessed February 16, 2026, https://arxiv.org/abs/2309.14030
18. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - Cloudfront.net, accessed February 16, 2026, https://ds8526jcpbygs.cloudfront.net/wp-content/uploads/2023/12/203_dewave_discrete_encoding_of_ee.pdf
19. NeurIPS Poster DeWave: Discrete Encoding of EEG Waves for EEG to Text Translation, accessed February 16, 2026, https://neurips.cc/virtual/2023/poster/71403
20. (PDF) The acoustics of stonehenge - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/283252397_The_acoustics_of_stonehenge
21. Resonances | Sounds of Stonehenge - WordPress.com, accessed February 16, 2026, https://soundsofstonehenge.wordpress.com/resonances/
22. Conclusions | Sounds of Stonehenge - WordPress.com, accessed February 16, 2026, https://soundsofstonehenge.wordpress.com/conclusions/
23. A Remarkable New Study Suggests That Stonehenge Was Built to Amplify Sound During Ancient Ruling-Class Rituals | Artnet News, accessed February 16, 2026, https://news.artnet.com/art-world/stonehenge-acoustics-1905935
24. Songs of the Stones - The acoustics of Stonehenge - pseudoscience? - www.Sarsen.org, accessed February 16, 2026, https://www.sarsen.org/2014/08/songs-of-stones-acoustics-of-stonehenge.html
25. 15.6 Technology Research - Bringing It All Together - Scribd, accessed February 16, 2026, https://www.scribd.com/document/947456153/15-6-Technology-Research-Bringing-It-All-Together

26. FallViewer: A Fine-grained Indoor Fall Detection System with Ubiquitous Wi-Fi Devices | Request PDF - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/349791362_FallViewer_A_Fine-grained_Indoor_Fall_Detection_System_with_Ubiquitous_Wi-Fi_Devices
27. Wireless Brain Sensors: A Revolutionary Leap in Neuroscience - DelveInsight, accessed February 16, 2026, https://www.delveinsight.com/blog/wireless-brain-sensors-in-healthcare
28. Snooping Keystrokes with mm-level Audio Ranging on a Single Phone - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/299866955_Snooping_Keystrokes_with_mm-level_Audio_Ranging_on_a_Single_Phone
29. Turning a Mobile Device into a Mouse in the Air | Request PDF - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/300664311_Turning_a_Mobile_Device_into_a_Mouse_in_the_Air
30. Context-free Attacks Using Keyboard Acoustic Emanations | Request PDF - ResearchGate, accessed February 16, 2026, https://www.researchgate.net/publication/288104632_Context-free_Attacks_Using_Keyboard_Acoustic_Emanations
31. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - arXiv, accessed February 16, 2026, https://arxiv.org/html/2309.14030v1
32. Unveiling Thoughts: A Review of Advancements in EEG Brain Signal Decoding into Text, accessed February 16, 2026, https://www.alphaxiv.org/overview/2405.00726v2
33. Ubiquitous WiFi and Acoustic Sensing: Principles, Technologies, and Applications, accessed February 16, 2026, https://jcst.ict.ac.cn/fileup/1000-9000/PDF/JCST-2023-1-3-3073-25.pdf
34. A Wearable Internet of Things-Based Device for the Quantitative Assessment of Hand Tremors in Parkinson's Disease: The ELENA Project - PMC, accessed February 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12074293/
35. DeWave: The AI That Can Accurately Convert Your Thought to Text - Analytics Vidhya, accessed February 16, 2026, https://www.analyticsvidhya.com/blog/2024/01/dewave-the-ai-that-can-accurately-convert-your-thoughts-into-text/
36. Portable, non-invasive, mind-reading AI turns thoughts into text - UTS, accessed February 16, 2026, https://www.uts.edu.au/news/2023/12/portable-non-invasive-mind-reading-ai-turns-thoughts-text
37. This AI Paper Unveils DeWave: Revolutionizing EEG-to-Text Translation with a Novel AI Framework for Open-Vocabulary BCIs - MarkTechPost, accessed February 16, 2026, https://www.marktechpost.com/2023/12/16/this-ai-paper-unveils-dewave-revolutionizing-eeg-to-text-translation-with-a-novel-ai-framework-for-open-vocabulary-bcis/